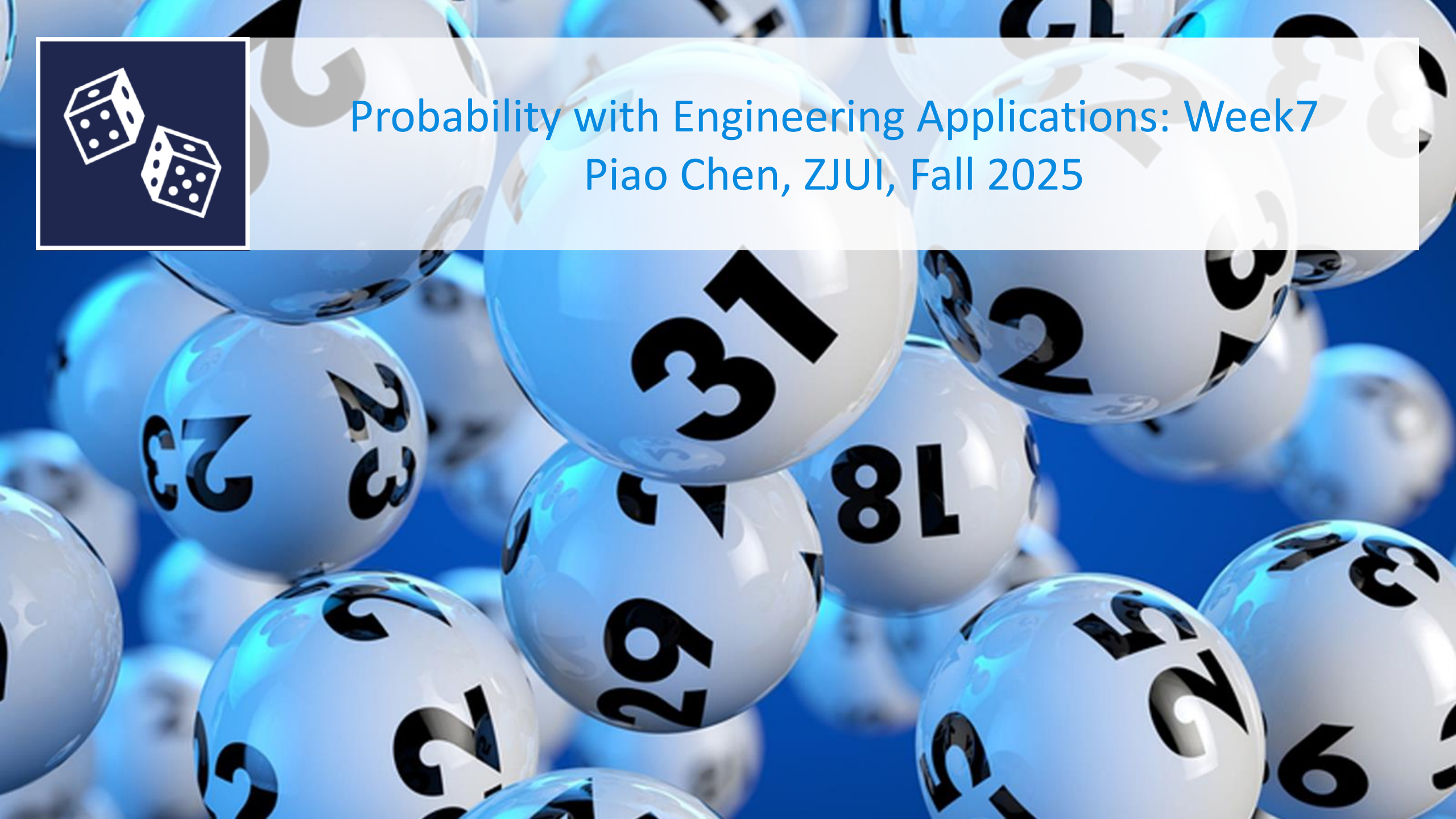




Probability with Engineering Applications: Week7

Piao Chen, ZJUI, Fall 2025



Programme

- Function of a random variable
- Generating a random variable

Function of a random variable

Function of a Random Variable

The above illustrates the **transformation** of a random variable

We will develop a general procedure to analyze such transformations, enabling us to:

- Determine $F_Y(c)$ given $F_X(c)$ and the transformation g
- Design the transformation g so that $F_Y(c)$ fits our needs

Three Step Process

Suppose X is a **continuous-type random variable** with pdf $f_X(u)$ and $Y = g(X)$ for some function g

What is $f_Y(u)$?

Steps find the transformed pdf:

1. **Scope the problem.** Plot g and consider the support of $f_X(u)$ and the range of g .
2. Find the CDF $F_Y(u) = P\{g(X) < u\}$
3. Differentiate to find $f_Y(u)$

Example 1

If $X \sim \text{Uniform}(-\frac{\pi}{2}, \frac{\pi}{2})$ and $Y = \tan(X)$, what is $f_Y(u)$?

1. Scope the problem.

Example 1 (continued)

2. Determine the CDF

Example 1 (continued)

3. Differentiate to determine the pdf.

Properties of the Cauchy Distribution

Does the proposed pdf integrate to 1?

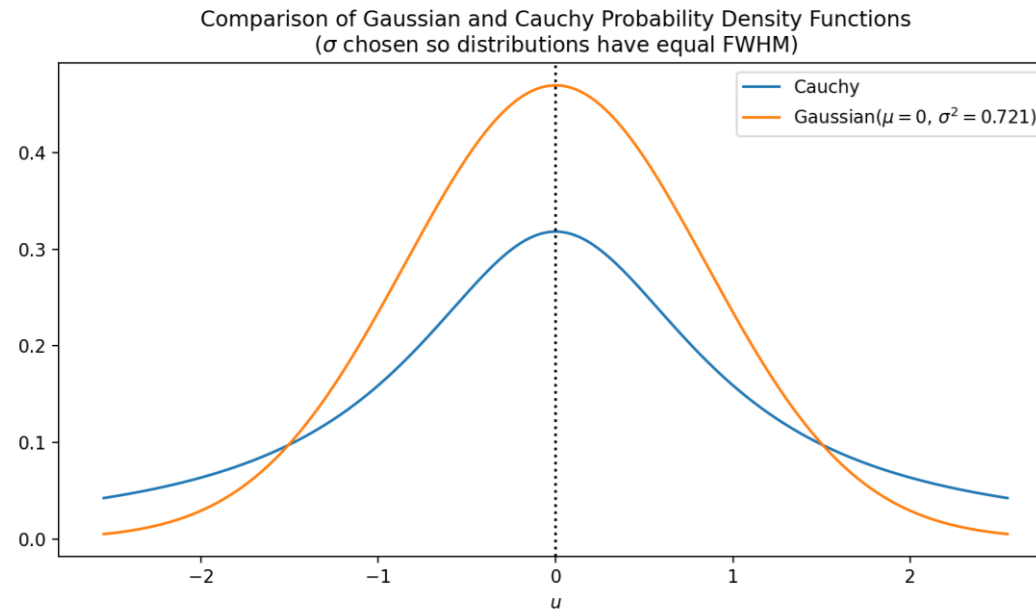
What is the median (value that splits the probability density in half)?

What is the mode (argmax of the pdf)?

Mean of Cauchy Distribution

What is $E[Y]$ if $Y \sim \text{Cauchy}$?

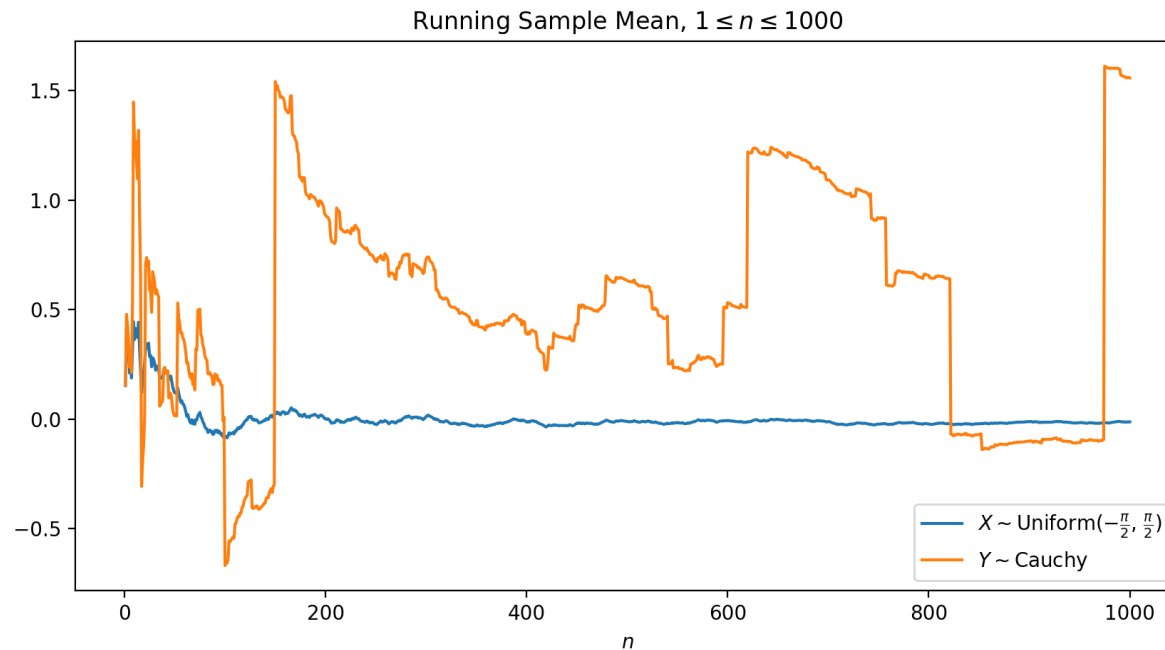
What is an Undefined Mean?



- The Gaussian probability density function is similar in shape
- The Cauchy distribution has much “heavier tails” (not exponentially bounded)

What is an Undefined Mean?

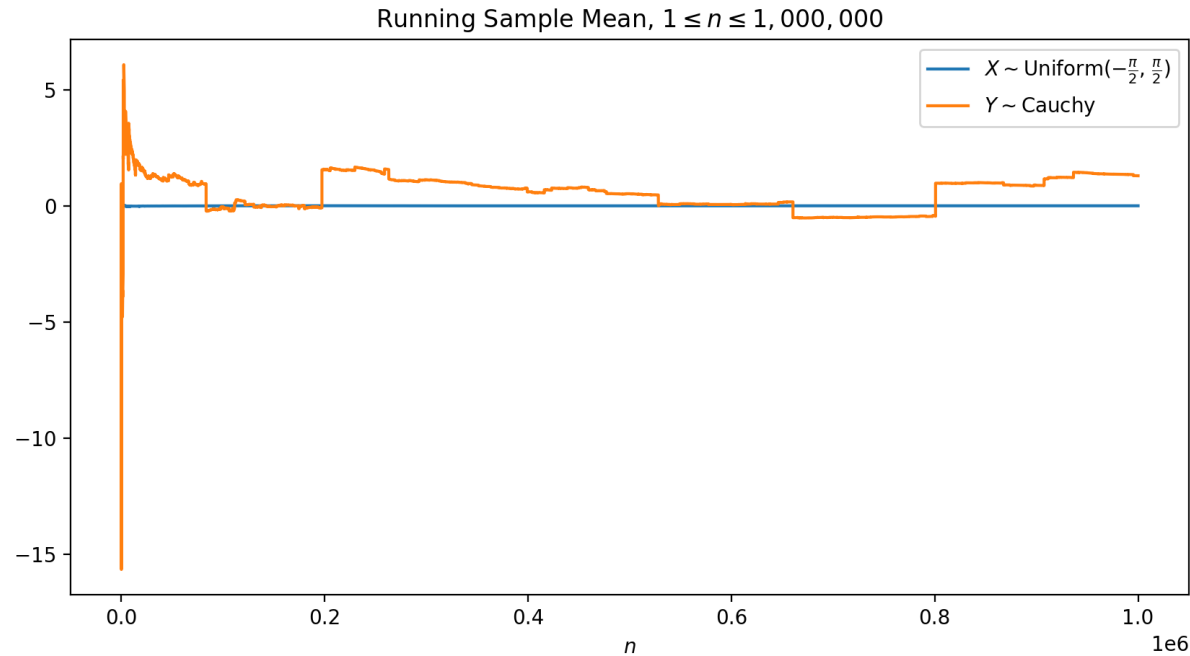
Consider the running sample mean: $\bar{x}(n) = \frac{1}{n} \sum_{i=1}^n x_i$



Note that $\bar{x}(n)$ (Uniform samples) settles around 0 while $\bar{y}(n)$ (Cauchy samples) is erratic

What is an Undefined Mean?

Consider the running sample mean: $\bar{x}(n) = \frac{1}{n} \sum_{i=1}^n x_i$



The erratic behavior of $\bar{y}(n)$ (Cauchy samples) persists as the number of samples n increases

Example 2

Let $X \sim \text{Exponential}(\lambda)$ and $Y = X - \text{floor}(X)$ where
$$\text{floor}(x) = \max\{m \in \mathbb{Z} \mid m \leq x\}$$

(the largest integer that is no larger than x)

1. Scope the problem.

Example 2 (continued)

2. Determine the CDF.

Example 2 (continued)

3) Differentiate to determine the pdf.

Transformation of a Random Variable: Continuous-Type

General Case

- Consider the transformation of a random variable: $Y = g(X)$
- Suppose:
 - Y is continuous-type
 - The inverse function g^{-1} exists
- A *sufficient* condition for g^{-1} to exist is if g is a strictly increasing function
 - Recall: g is strictly increasing if $g(x) < g(y) \quad \forall \quad x < y$
- The CDF of Y can be determined:

$$F_Y(c) = P\{Y \leq c\} = P\{X \leq g^{-1}(c)\} = F_X[g^{-1}(c)]$$

Transformation of a Random Variable: Continuous-Type

General Case

The pdf can be determined by differentiating the CDF

Transformation of a Random Variable: Continuous-Type

General Case

Example: What is the derivative of $g^{-1}(x) = \arctan(x)$?

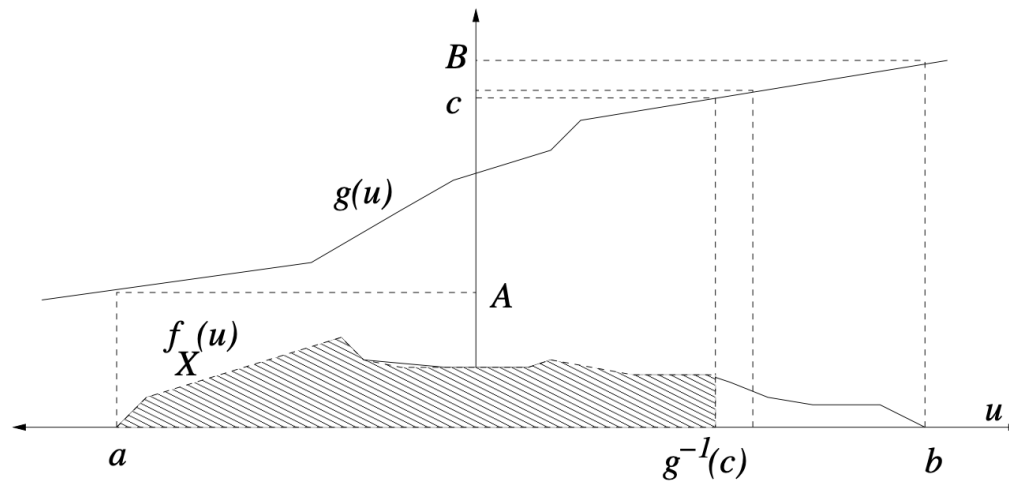
Transformation of a Random Variable: Continuous-Type

General Case

Suppose f_X is nonzero on the interval $[a, b]$

Let $A = g(a)$ and $B = g(b)$. Then

$$f_Y(u) = \begin{cases} f_X[g^{-1}(u)] \frac{1}{g'[g^{-1}(u)]}, & A < u < B \\ 0, & \text{otherwise} \end{cases}$$



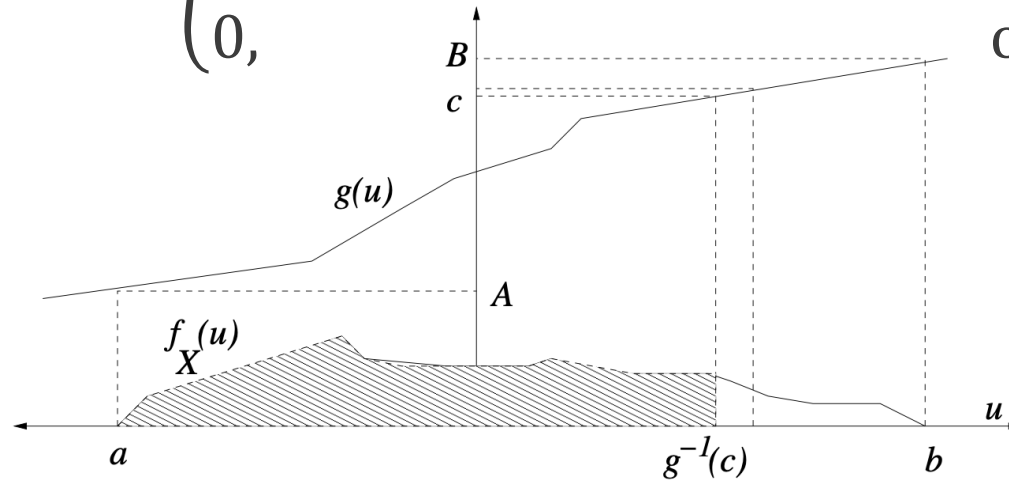
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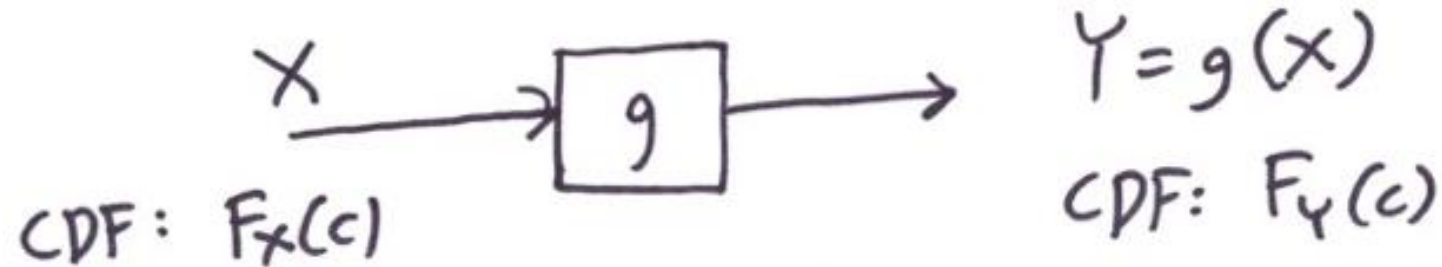
Revisit Cauchy Example

If $X \sim \text{Uniform}(-\frac{\pi}{2}, \frac{\pi}{2})$ and $Y = \tan(X)$, what is $f_Y(u)$?

Example: Lognormal Distribution

Suppose $X \sim N(\mu, \sigma^2)$. Then the random variable $Y = e^X$ is said to be a **lognormal** random variable. What is the pdf of Y ?

Transformation of Random Variable: Discrete-Type Case

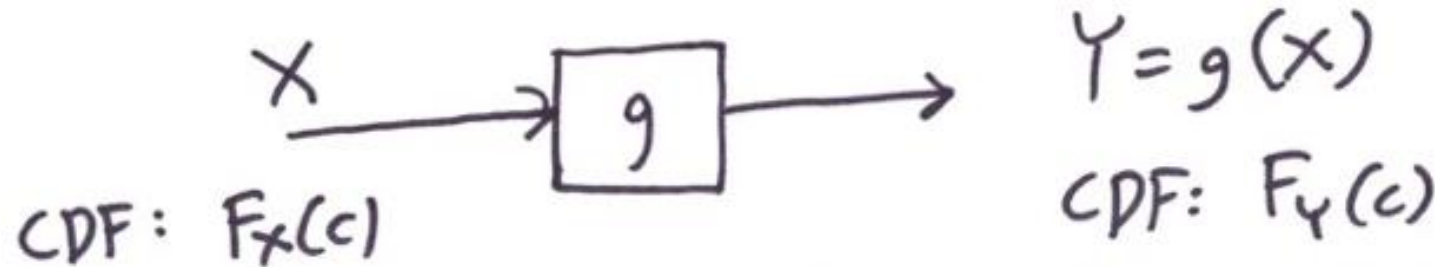


If g is a piecewise constant function, then Y will be a discrete-type random variable

Two step process to determine the pmf:

1. **Scope the problem.** Plot g and consider the support of $p_X(u)$ and the range of g .
2. **Find the pmf.** If X is discrete-type:

Transformation of Random Variable: Discrete-Type Case



If g is a piecewise constant function, then Y will be a discrete-type random variable

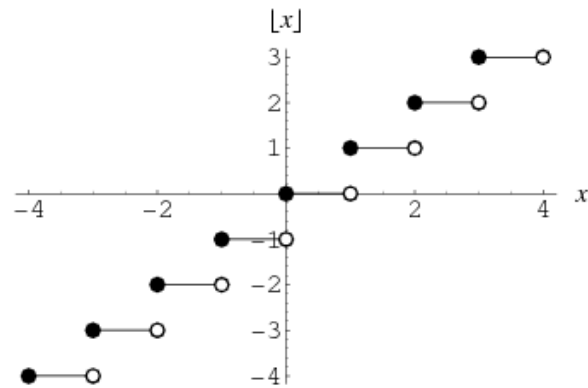
Two step process to determine the pmf:

1. **Scope the problem.** Plot g and consider the support of $p_X(u)$ and the range of g .
2. **Find the pmf.** If X is continuous type:

Example 1

Suppose $X \sim \text{Exp}(\lambda)$ and $Y = \text{floor}(X)$. What is the pmf of Y ?

1. Scope the problem



The function g is piecewise constant $\Rightarrow Y$ is discrete-type

The function g maps real numbers $(-\infty, \infty)$ to the integers

Example 1

2. Determine the pmf.

Example 2 (Hajek 3.28)

Suppose X has pdf $f_X(u) = \frac{e^{-|u|}}{2}$ and $Y = g(X)$ where

$$g(u) = \alpha \operatorname{sign}(u) = \begin{cases} \alpha, & u \geq 0 \\ -\alpha, & u < 0 \end{cases}$$

for some positive constant α .

In other words: Y is the output of a binary quantizer with input X .

a) What is the pdf or pmf of Y ?

Example 2 (Hajek 3.28)

b) Find the mean square quantization error $E[(X - Y)^2]$

Example 2 (Hajek 3.28)

c) Find α to minimize the mean square quantization error

Generating a random variable

Example 3.8.11 Suppose X is a continuous-type random variable with CDF F_X . Let Y be the result of applying F_X to X , that is, $Y = F_X(X)$. Find the distribution of Y .



Example 1

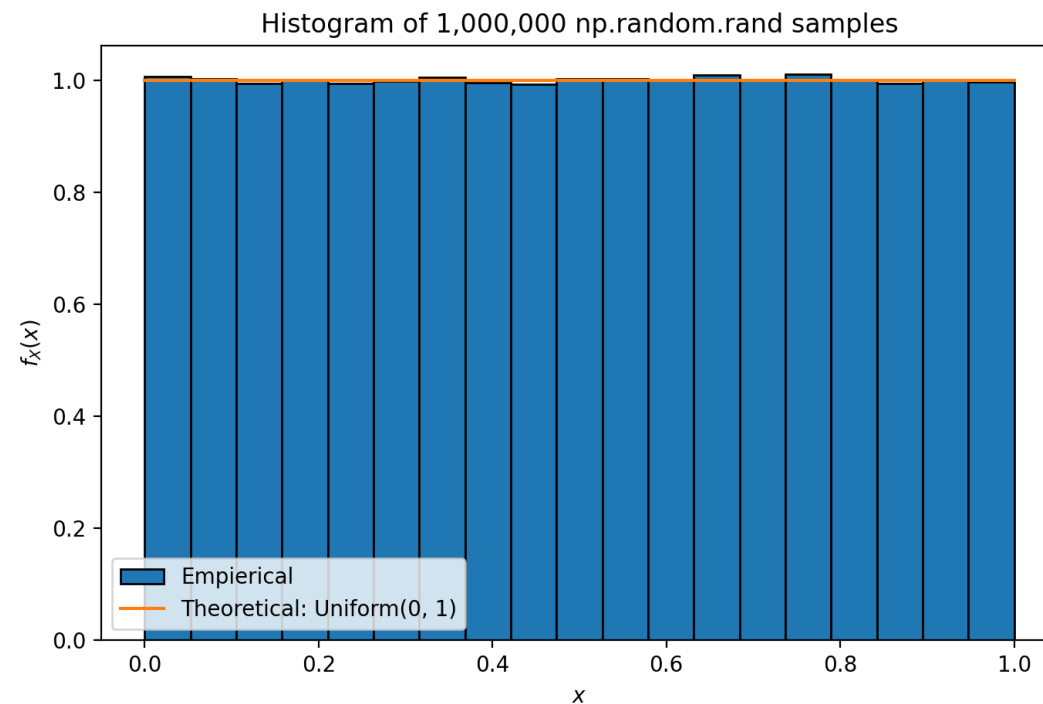
Suppose you can generate $X \sim \text{Uniform}(0, 1)$ on the computer (even the C standard library includes the rand function)

What transformation $Y = g(X)$ produces $Y \sim \text{Exponential}(\lambda)$?



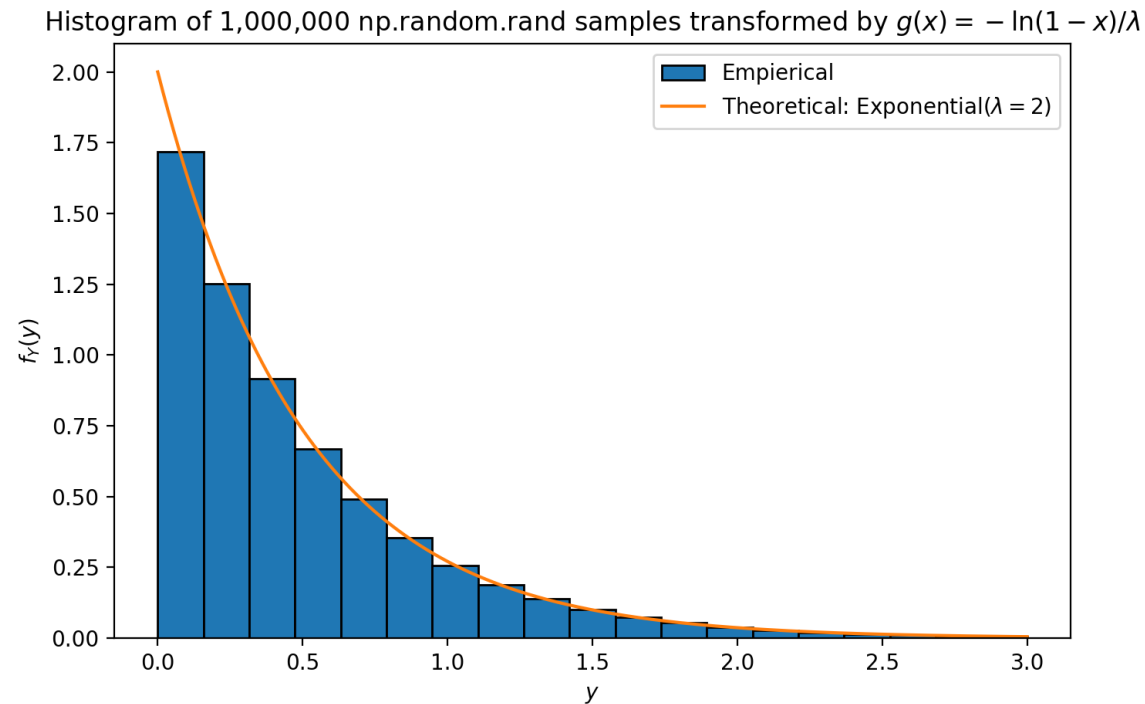
Example 1

Consider 10^6 Uniform(0, 1) random samples calculated with the NumPy random.rand function



Example 1

Consider 10^6 Uniform(0, 1) random samples transformed by the function $g(x) = -\ln(1 - x)/\lambda$



The area rule for expectation based on the CDF





See you next week